

Machine Learning (Stanford CS229)

Stanford Frontier AI Learning System

Baseline: October 6, 2026. Print companion to the HTML course site:
crash course and cheatsheet.

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Crash course: the whole course in thirty minutes

Seventeen units, under two minutes each. Every worked number below comes from the named lesson section. The crash course points, the lessons prove.

U01, Learning problems, risk, setup (lessons/u01)

A learning problem is data, a hypothesis family, a loss, and a risk. Empirical risk is the average loss on the training set. Population risk is the average over the true distribution. You optimize the first and care about the second. The gap between them is generalization.

Worked number (U01): $m = 2$ patients, $n = 3$ features. X has shape $(2, 3)$: 2 examples, 3 dimensions. $\theta = [1, 1]$ gives $h(x) = 1 + x$. On $x = 1$, $h = 2$, $y = 2$, error 0.

[!MEMORY] Risk is the goal. The training set is the evidence. Never confuse them.

Object	Notation	Meaning
Features	x in $\mathbb{R}^n$	The inputs
Target	y	The label
Hypothesis	h_θ	The predictor
Empirical risk	$(1/m) \sum \text{loss}$	Training average
Population risk	$E[\text{loss}]$	True average

Self-test:

[!QA] Q: Why is low training error not enough? A: Training error is empirical risk. Deployment pays population risk. The gap is generalization, and U08 measures it.

U02, Linear regression (lessons/u02)

Fit $h(x) = \theta^T x$ by minimizing squared error. Three ways: stochastic gradient (one example at a time), batch gradient (all examples), or the normal equations (one linear solve). Gaussian noise makes least squares the MLE. Locally weighted regression fits a new linear model at each query point: nonparametric, no fixed θ .

Worked number (U02): $\theta = [0, 0]$, $\alpha = 0.1$, first stochastic step on $(x = 1, y = 2)$: $h = 0$, error 2, $\theta := [0.2, 0.2]$. Normal equations on $X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$, $y = [2, 3, 4]$: $X^T X = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$, $X^T y = [9, 20]$, solve gives $\theta = [1, 1]$.

[!MEMORY] Gradient steps are cheap and many. The normal equations are one shot and cubic.

Self-test:

[!QA] Q: When do you pick the normal equations over gradient descent? A: Small d , well-conditioned $X^T X$, and you want the exact answer. For $n = 10^7$ and $d = 100$, gradient descent wins on memory.

U03, Classification and GLMs (lessons/u03)

Logistic regression models $p(y = 1 | x) = \text{sigmoid}(\theta^T x)$. The gradient is $(y - h) x$: the same error-times-feature shape as LMS. GLMs generalize this: pick an exponential-family noise model, its canonical link gives the predictor. Softmax is the multiclass link.

Worked number (U03): $\theta = [-4, 1]$, $x = [1, 3]$, $y = 1$. $h = 0.2689$, error 0.7311, $\alpha = 0.5$: $\theta := [-3.6344, 2.0967]$. New $h = 0.9344$. One step moves the probability from 0.27 to 0.93.

[!MEMORY] Sigmoid in, cross-entropy out. The gradient is always error times feature.

Self-test:

[!QA] Q: The data is linearly separable and the weights blow up. What happened? A: The MLE is at infinity on separable data. Regularize or stop early.

U04, Generative classifiers (lessons/u04)

Model $p(x | y)$ and $p(y)$, then use Bayes rule. GDA fits a Gaussian per class with shared covariance: the boundary is linear. Naive Bayes assumes features independent given the class. Discriminative models (logistic) usually need less data when the Gaussian assumption is wrong, generative models win when it is right.

Worked number (U04): $\Sigma = \begin{bmatrix} 1 & 0.8 \\ 0.8 & 1 \end{bmatrix}$, determinant 0.36. At $x = [1, 1]$, the Mahalanobis distance uses $\Sigma^{-1} = \begin{bmatrix} 2.778 & -2.222 \\ -2.222 & 2.778 \end{bmatrix}$.

[!MEMORY] Generative models the inputs. Discriminative models the boundary.

Self-test:

[!QA] Q: GDA and logistic regression both give linear boundaries. When does GDA win? A: When the Gaussian assumption holds and data is scarce: GDA uses the input distribution, logistic does not.

U05, Kernels (lessons/u05)

A kernel $K(x, z) = \langle \phi(x), \phi(z) \rangle$ computes an inner product in a rich feature space without building ϕ . The representer theorem: the optimal solution lives in the span of the training points. Cost: the $n \times n$ Gram matrix.

Worked number (U05): $x = [1, 2]$, $z = [3, 1]$, $\langle x, z \rangle = 5$. Degree-2 polynomial kernel: $(1 + 5)^2 = 36$. Same as the explicit inner product in the degree-2 feature space.

[!MEMORY] The kernel is a shortcut. The Gram matrix is the bill.

Self-test:

[!QA] Q: Why can a kernel beat explicit features? A: Infinite-dimensional ϕ (like RBF) has no explicit finite form. The kernel computes the inner product anyway.

U06, SVMs (lessons/u06)

Maximize the geometric margin: the distance from the boundary to the nearest point. The dual depends only on inner products, so kernels plug in. Only support vectors matter. C trades margin width against violations. SMO solves the dual two variables at a time.

Worked number (U06): $w = [3,4]$, $b = -25$, $x = (4,3)$, $y = 1$. Score -1, functional margin -1, $\|w\| = 5$, geometric margin -0.2: the point is 0.2 units on the wrong side. Doubling (w, b) doubles the functional margin but the geometric margin stays -0.2.

[!MEMORY] Functional margin scales. Geometric margin does not. Optimize the geometric one.

Self-test:

[!QA] Q: All alphas hit the C bound. What does that tell you? A: C is too small or the data is very noisy: every point wants to be a margin violator.

U07, Neural nets and backprop (lessons/u07)

A network is a composition of modules. Each module's backward contract: given dL/dy , return dL/dx and dL/dW . The chain rule does the rest. Activations must be nonlinear or the depth collapses to one linear map. Check every implementation with finite differences.

Worked number (U07): $z = [1,2,3]$, LayerNorm gives $[-1.2247, 0, 1.2247]$. Scale by 5 first: identical output. The normalization is scale-invariant.

[!MEMORY] Forward is composition. Backward is the chain rule. Test with finite differences.

Self-test:

[!QA] Q: Gradients are exactly zero below layer 3. Name two causes. A: Dead ReLUs everywhere below, or a detached graph (no_grad or wrong wiring).

U08, Generalization (lessons/u08)

Test error = $\text{bias}^2 + \text{variance} + \text{noise}$. Simple models: high bias, low variance. Complex models: low bias, high variance. Finite hypothesis classes: union bound gives sample complexity $\log|H|$. Double descent: past the interpolation threshold, test error falls again.

Worked number (U08): truth x^2 on $[0,1]$, noise 0.3, $n = 15$. Linear: bias^2 0.05, variance 0.002. Degree-5: bias^2 0, variance 0.08. The quadratic hits bias^2 0, variance 0.008: the sweet spot.

[!MEMORY] Bias is what the family cannot fit. Variance is what the data cannot pin down.

Self-test:

[!QA] Q: Test error rises while train error falls. Two causes, two fixes? A: Overfitting: regularize or stop early. Distribution shift: fix the data pipeline, not the training loop.

U09, Regularization and model selection (lessons/u09)

Ridge (l2) shrinks weights, LASSO (l1) zeroes some. Early stopping is implicit regularization. Bayesian view: MAP with a prior. Choose hyperparameters by cross-validation, never on the test set. Leakage is the silent killer: any test information in training invalidates the estimate.

Worked number (U09): degree-8 polynomial, $n = 12$. Test MSE: unregularized 2.41, ridge $\lambda = 1$ gives 0.31, $\lambda = 100$ gives 0.62. The sweet spot is between.

[!MEMORY] Regularization is a prior. Cross-validation is the judge. Leakage is the crime.

Self-test:

[!QA] Q: You tuned on the test set and report 0.95 accuracy. What is wrong? A: The test set is now a validation set. The reported number is optimistic. Get a fresh test set.

U10, Clustering and EM (lessons/u10)

k-means alternates: assign points to the nearest center, recompute centers. It converges to a local minimum, not the global one. EM generalizes this to mixtures: the E step computes soft responsibilities, the M step maximizes the expected complete likelihood. Jensen's inequality builds the lower bound. Likelihood never decreases.

Worked number (U10): two Gaussians, $n = 200$, $k = 2$. k-means objective falls 7.96 to 0.98 in 4 iterations, monotone. With $k = 3$ the objective is lower but the third cluster splits a real one: lower objective is not truer clusters.

[!MEMORY] k-means is hard EM with spherical Gaussians. EM never decreases the likelihood.

Self-test:

[!QA] Q: EM converged but the clusters are meaningless. What do you check? A: Initialization (multiple restarts), k (model selection), and whether the likelihood gain came from a variance collapsing to zero.

U11, VI, PCA, ICA (lessons/u11)

PCA keeps the directions of maximum variance: eigendecomposition of the covariance, or SVD of the centered data. ICA separates independent non-Gaussian sources: independence is stronger than decorrelation. Variational inference maximizes the ELBO when the posterior is intractable. The VAE is VI with neural nets.

Worked number (U11): 1000 points near a plane in 3d. Eigenvalues 2.26, 1.62, 0.04. Top-2 PCA keeps 98.9% of the variance.

[!MEMORY] PCA decorrelates. ICA independizes. VI bounds what it cannot compute.

Self-test:

[!QA] Q: PCA gives uncorrelated components. Why is that not enough for source separation?

A: Uncorrelated is second-order. Independence needs higher orders. Gaussian sources cannot be separated at all.

U12, Diffusion (lessons/u12)

Corrupt data with a fixed Gaussian schedule, then learn to reverse it. The ELBO becomes a denoising objective: predict the noise (or the clean x_0 , or the score: three parameterizations, one model). Sampling runs the reverse chain. The schedule is load-bearing: α -bar must reach 0 exactly at T .

Worked number (U12): $T = 100$, beta linear $1e-4$ to 0.02 , $x_0 = 3.0$. $\alpha\text{-bar}_{50} = 0.7772$, so x_{50} has mean 2.6447 and variance 0.2228 .

[!MEMORY] Forward is fixed corruption. Reverse is learned denoising. The schedule is the contract.

Self-test:

[!QA] Q: Samples are blurry gray blobs. Two possible causes? A: The reverse net predicts the mean (underfit), or the schedule destroys signal too early.

U13, Foundation models (lessons/u13)

Pretrain once on broad data, adapt cheaply: linear probing (frozen backbone), full fine-tuning, or LoRA (low-rank adapters). Contrastive learning pulls positives together and pushes negatives apart: the InfoNCE loss. RAG grounds generation in retrieved documents.

Worked number (U13): $d = 4096$, $r = 8$. Dense update: 16.8M parameters. LoRA: 65,536 parameters, a 256x reduction.

[!MEMORY] Pretraining buys the representation. Adaptation rents a small corner of it.

Self-test:

[!QA] Q: Linear probing fails but full fine-tuning works. What does that tell you? A: The pre-trained features do not linearly separate your task. The representation needs reshaping, not just a new head.

U14, LLMs (lessons/u14)

Tokens in, next-token loss, transformer blocks. The causal mask enforces autoregression: position t sees only the past. KV cache makes generation linear per step instead of quadratic. GQA and MQA shrink the cache. SFT trains on answer tokens only (the loss mask). Temperature and top-p are inference choices, not training choices.

Worked number (U14): $d = 4096$, 32 heads, fp16, $T = 8192$. KV cache per layer: MHA 128 MiB, GQA (8 groups) 32 MiB, MQA 4 MiB. The cache scales with the number of KV heads.

[!MEMORY] The mask is the autoregression. The cache is the memory bill. The loss mask is the SFT detail.

Self-test:

[!QA] Q: Training loss is near zero after one epoch but generation is incoherent. Diagnose. A: The causal mask has a gap or sits in the wrong place: the model reads the future during training.

U15, Reasoning and RL foundations (lessons/u15)

Chain of thought turns one hard prediction into many small ones. RLVR trains traces with verifier rewards: the reward is terminal, the KL penalty guards drift, GRPO replaces the critic with group-relative advantages. Underneath: MDPs, Bellman equations, value and policy iteration, model learning from counts, and continuous-state methods (discretization vs value approximation).

Worked number (U15): two-state toy, $\gamma = 0.9$. $V(B) = 10$, $V(A) = 9$. Value iteration error ratio exactly 0.9 per sweep: the Bellman operator is a γ -contraction. GRPO on rewards $[1, 0, 0, 1]$: advantages $[1, -1, -1, 1]$.

[!MEMORY] The verifier is the objective. The Bellman equation is the recursion. The error shrinks by γ .

Self-test:

[!QA] Q: RLVR reward rises but task accuracy is flat. Diagnose. A: The verifier is invalid: it rewards a proxy (format, length), not the task. Audit the verifier, not the model.

U16, Control and policy optimization (lessons/u16)

Finite-horizon MDPs solve backward with non-stationary policies. LQR: linear dynamics plus quadratic costs give a closed-form Riccati solution with a linear policy that ignores the noise. DDP iterates linearizations along a trajectory. LQG adds the Kalman filter for partial observation. REINFORCE: the log-derivative trick, baselines cut variance, PPO clips the likelihood ratio to stay near the old policy.

Worked number (U16): 1d LQR, $A = B = U = W = 1$, $T = 2$. $\Phi = [-1, -1.5, -1.6]$, gains $L = [-0.5, -0.6]$. From $s_0 = 5$: $5 \rightarrow 2 \rightarrow 1$. Kalman 1d step: $K = 2/3$, estimate 1.0, variance $2/3$. PPO clip, $\epsilon = 0.2$: $(1, 1.3) \rightarrow 1.2$ capped, $(-1, 0.5) \rightarrow -0.8$ capped.

[!MEMORY] Riccati goes backward. The Kalman gain weights the innovation. The clip bounds the step.

Self-test:

[!QA] Q: The LQR closed loop diverges although the recursion ran. Two causes? A: The gain sign is flipped (the notes print it unsigned), or W is not positive definite.

U17, Appendices and synthesis (lessons/u17)

The four identities: Gaussian sums stay Gaussian, Gaussian conditioning, same-covariance KL, the KL chain rule. The historical supplements (factor analysis, online learning, HMMs, GPs) live on inside EM, Kalman, and kernels. Research craft: baselines with tests, one-factor ablations with matched budgets, honest error bars, and the deployment gap map.

Worked number (U17): ten seeds, returns around 8.1. Mean 8.10, sample std 0.18, standard error 0.058. Bootstrap 95% interval $[7.99, 8.21]$. $KL(N([1,0],I) || N([0,0],I)) = 0.5$, Monte Carlo 0.4998.

[!MEMORY] Report all seeds. One factor per ablation. A difference inside the error bars is not a win.

Self-test:

[!QA] Q: Your method beats the baseline 0.83 to 0.79. Is it better? A: Not established: no spread, no sample size, no budget statement. Overlapping error bars mean noise.

The whole course in one table

Unit	Core mechanism	Key number
U01	Empirical vs population risk	X shape (2,3)
U02	Least squares, 3 solvers	theta [0.2,0.2] after 1 SGD step
U03	Sigmoid, GLMs	h 0.27 -> 0.93 in 1 step
U04	GDA, naive Bayes	det Sigma 0.36
U05	Kernel trick	$(1+5)^2 = 36$
U06	Max margin, dual	geometric margin -0.2
U07	Backprop chain rule	LayerNorm scale-invariant
U08	Bias/variance	quadratic wins: var 0.008
U09	Ridge, CV	MSE 2.41 -> 0.31
U10	k-means, EM	J 7.96 -> 0.98
U11	PCA/ICA/VI	98.9% variance in top-2
U12	Denoising objective	alpha-bar_50 0.7772
U13	Pretrain + adapt	LoRA 256x fewer params
U14	Causal attention, KV cache	MQA 4 MiB vs MHA 128 MiB
U15	Bellman, RLVR	error ratio 0.9/sweep
U16	Riccati, PPO	5 -> 2 -> 1 trajectory
U17	KL identities, craft	se 0.058 over 10 seeds

Final self-test

[!QA] Q: Name the assumption each of these breaks: naive Bayes on correlated features, LQR on a swing-up, PPO with 50 epochs per batch. A: Conditional independence, linearity near one fixed point, the local surrogate (the state distribution shifted).

Cheatsheet: formulas, shapes, traps, decision rules

One dense page for the whole course. Formulas use the notation fixed in notation and shapes. Section pointers name the lesson that proves each line.

U01, learning problems and risk

Formula / shape	Trap	Decision rule
$X: m \times n$, m examples, n features	Transposed X flips examples and features	Say "m by n" out loud before you code
Empirical risk: $(1/m)$ sum loss	Optimizing it is not the goal	Watch the gap, not the train loss
Population risk: $E[\text{loss}]$	Unobservable directly	Estimate with a clean test set

U02, linear regression

Formula / shape	Trap	Decision rule
$\theta := \theta - \alpha X^T(X\theta - y)/m$	α too large diverges	Sweep α on a log grid
$\theta = (X^T X)^{-1} X^T y$	$X^T X$ singular or ill-conditioned	Check cond before you invert
Gaussian noise $\Rightarrow$ least squares is MLE	Real noise is not Gaussian	Treat it as a modeling choice
Local weighting: fit per query	$O(n)$ memory and $O(n)$ per query	Use for small n only

U03, classification and GLMs

Formula / shape	Trap	Decision rule
$h = 1/(1 + e^{-\theta^T x})$ $\text{grad} = (y - h) x$	Overflow at extreme logits Same shape as LMS, different h	Use the log-sum-exp stable form Check the link before the gradient
Softmax: $e^{z_i} / \sum e^{z_j}$	Forgetting the normalization	Rows must sum to 1
Separable data $\Rightarrow \ \theta \ \rightarrow \infty$	Unregularized training never converges	Add L_2 or stop early

U04, generative classifiers

Formula / shape	Trap	Decision rule
$p(y x) = p(x y)p(y)/p(x)$	Wrong prior ruins the posterior	Estimate ϕ from data, do not guess
GDA shared Sigma => linear boundary	Separate covariances => quadratic	Test which fits with CV
Naive Bayes: features indep	Correlated features double-count	Check correlation first
Laplace smoothing	Zero counts kill products	Smooth before you multiply

U05, kernels

Formula / shape	Trap	Decision rule
$K(x,z) = \langle \phi(x), \phi(z) \rangle$	Kernel must be PSD	Check eigenvalues on a subset
Gram matrix: $n \times n$	$O(n^2)$ memory	1M points = 8 TB: do not build it
Representer: $w = \sum \alpha_i \phi(x_i)$	Solution needs all n alphas	Sparsity comes from the loss, not free

U06, SVMs

Formula / shape	Trap	Decision rule
Geometric margin = $y(w^T x + b) / \ w\ $	Functional margin scales freely	Constrain $\ w\ $, then maximize
Dual in α_i , $0 \leq \alpha_i \leq C$	All α at C means C too small	Raise C or accept noise
SMO: 2 variables at a time	Kernel cache thrash	Keep the Gram rows you reuse
Support vectors define the boundary	Outliers become support vectors	Inspect them, do not worship them

U07, neural nets

Formula / shape	Trap	Decision rule
dL/dW from dL/dY via chain rule	Shape mismatch in batched code	Write shapes on every line
ReLU: $\max(0, z)$	Dead units give zero gradient	Monitor fraction of active units
Init: Var-preserving scale	Wrong scale => vanish/explode	Match init to the activation
Finite differences check	Too large epsilon lies	Use $1e-5$ relative error below $1e-5$

U08, generalization

Formula / shape	Trap	Decision rule
Err = bias ² + var + noise	One number hides the split	Plot learning curves
Union bound: $\log H /m$	Vacuous for huge H	Use it for intuition, not guarantees
Double descent peak at $d = n$	Tuned-away peak looks like no peak	Report the lambda you used

U09, regularization

Formula / shape	Trap	Decision rule
Ridge: $(X^T X + \lambda I)^{-1} X^T y$	lambda by habit	Choose by CV
LASSO zeroes coordinates	Exact zeros need coordinate descent	Do not expect zeros from SGD
CV error estimates test error	Tuning on test leaks	Lock the test set until the end
Early stopping	Stopping rule is a hyperparameter	Fix it before you compare

U10, clustering and EM

Formula / shape	Trap	Decision rule
k-means: assign, recompute	Local minima	Restart many times
EM: E responsibilities, M maximize	Variance collapse => inf likelihood	Floor the variances
Likelihood never decreases	"Never decreases" is not "converges to MLE"	It finds a local max

U11, PCA, ICA, VI

Formula / shape	Trap	Decision rule
PCA: eig of centered covariance	Forgetting to center	Center first, always
Keep k with 95%+ variance	Variance kept is not task signal	Validate downstream
ICA needs non-Gaussian sources	Gaussian mixtures cannot separate	Test non-Gaussianity first
ELBO $\leq \log p(x)$	Loose bound, good samples	Report the bound gap when you can

U12, diffusion

Formula / shape	Trap	Decision rule
$x_t = \sqrt{\alpha_{\bar{t}}} x_0 + \sqrt{1-\alpha_{\bar{t}}} \epsilon$	Schedule bug kills the signal	Assert $\alpha_{\bar{T}} = 0$
Denoising loss: $\ \epsilon - \epsilon_{\theta}\ ^2$	Predicting x_0 vs ϵ changes optimization	Pick ϵ -pred by default
Reverse: $p(x_{t-1} x_t)$ Gaussian	Wrong variance schedule	Use the fixed posterior variance

U13, foundation models

Formula / shape	Trap	Decision rule
LoRA: $W + B A, r \ll d$	r too small starves adaptation	Sweep r in $\{4, 8, 16\}$
InfoNCE: $-\log e^{\{s\}} / \sum e^{\{s\}}$	Easy negatives teach nothing	Mine hard negatives
Linear probe accuracy	Probes the features, not the task	Compare with full fine-tune
RAG: retrieve then generate	Retrieval errors become answer errors	Evaluate retrieval separately

U14, LLMs

Formula / shape	Trap	Decision rule
Attention: $\text{softmax}(QK^T / \sqrt{d_h} + M)V$	Mask after softmax leaks	Add the mask before softmax
KV cache: $2 T d$ bytes per layer	Provisioned for max T always	Account per actual sequence length
SFT loss mask on answers only	Prompt tokens in the loss	Mask them out
Temperature/top-p at inference	Tuning them like training knobs	Fix before you evaluate

U15, reasoning and RL

Formula / shape	Trap	Decision rule
$V^{\pi}(s) = R + \gamma E[V^{\pi}(s')]$	$\gamma \geq 1$ diverges	Keep γ in $[0, 1)$
Value iteration: $\max$ Bellman backup	Async order stalls	Check the residual
RLVR reward from a verifier	Verifier games the metric	Audit the verifier independently
GRPO advantage: $(R - \text{mean})/\text{std}$	Zero group variance $\Rightarrow$ no signal	Monitor group diversity

U16, control and policy optimization

Formula / shape	Trap	Decision rule
LQR gain: $-(B^T \Phi B - W)^{-1} B^T \Phi A$	The notes print it unsigned	Check Φ negative semidefinite
Riccati backward from $\Phi_T = -U_T$	W not positive definite	Check eigenvalues first
Kalman gain $K = P C^T (C P C^T + R)^{-1}$	Wrong noise scale	Calibrate from data
PPO clip: $\min(r A, \text{clip}(r) A)$	Many epochs on stale data	Watch KL and clipped fraction
Baseline keeps PG unbiased	Action-dependent baseline biases	Baseline on state only

U17, identities and craft

Formula / shape	Trap	Decision rule
KL same-cov Gaussian: d_{Maha}^2	Different covariances	Use the full formula
KL chain rule: split joint KL	Uninformative factorization	Pick the factorization that localizes
$se = s/\sqrt{n}$	$n = 3$ treated as Gaussian	Use the t-multiplier
One-factor ablation	Five changes at once	Change one thing per run
Acceptance gate: baseline, canary, rollback, owner	Shipping the training script	No gate, no deploy